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Domain Generalization of Deep Learning Models Under Subgroup Shift in Breast Cancer Prognosis

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***Abstract*—Making breast cancer prognosis from gene expression profiles of the primary tumor has become a promising application of deep learning. Yet, to be relevant to real world applications in the clinic and for knowledge discovery, these models must be robust to common distribution shifts. In this study, we evaluate recently proposed methods for improving domain and subgroup shifts. We test the in-distribution and out-of-distribution generalization of multiple episode learning, stochastic weight averaging, group distributionally robust optimization, and a subsampling scheme on one training and four external breast cancer prognosis datasets. The evaluation found that the methods can, to various degrees, improve generalization across domains, although there remain, partially high, generalization gaps. Additionally, in-distribution and out-of-distribution generalization differs between clinical subtypes of breast cancer. Thus, we conclude that further research into methods specifically addressing challenges in breast cancer prognosis from gene expression data are warranted.**

***Index Terms*—domain generalization, subgroup shift, breast cancer, survival analysis, gene expression**

I. Introduction

Despite the overall relatively high 5-year survival rate of breast cancer patients of around 90%, a large number of patients still succumb to the disease due to its high and increasing prevalence [1]. Breast cancer is a heterogeneous disease with different mechanisms of tumorgenesis, progression, and therapy resistance, among other factors. This is reflected in high variation of prognosis for different patients.

Increasingly, deep neural networks (DNNs) are trained to make personalized predictions about a patient's prognosis utilizing gene expression signatures from primary breast cancer tumors. These survival models allow assigning a scalar value to each patient representing their risk of death over time, also termed hazard [2]. Thus, patients that are at high risk of dying can be identified, potentially avoiding over- and undertreatment.

To be relevant in clinical practice and for biological knowledge discovery, trained DNN survival models are required to be robust to shifts in the distribution from which patient data are sampled. Most models are trained under the assumption that the training and test sets are sampled independently and identically from the same distribution (IID). However, in practice this assumption is regularly violated where the training and the test data are sampled from different distributions, termed out-of-distribution (OOD). This may lead to degradation of the models' performance [3].

Recent research has shown that overparameterized DNNs tend to learn compressed representations relating few features to the target variable, which seems to be beneficial for generalizing in the IID setting [4]. If spurious correlations have been learned, it becomes difficult for gradient descent-based algorithms to find additional features related to the target since gradient contributions in the direction of these features are small, as they do not significantly improve the loss [5]. However, if these spurious correlations are idiosyncratic to the training and internal test datasets, the trained model may perform poorly on an external dataset even if the underlying causal relations remain the same between domains [6]. This drawback of DNNs has led to extensive research into methods that increase robustness of DNNs to different types of distribution shifts [3]. However, it is not clear how such methods perform when applied to cancer prognosis based on transcriptomic data. In critical applications such as healthcare, reliance on spurious correlations and the therefrom resulting poor domain generalization may have catastrophic consequences. Addressing robustness of DNN-based survival models is therefore crucial.

In the context of breast cancer, we differentiate between two types of shifts in the underlying distribution. Firstly, domain shifts occur between datasets that stem from factors

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such as differences in specimen conservation method (fresh tissue [FT], fresh frozen [FF] or formalin-fixed paraffin-embedded [FFPE]), follow up time, genetic background, and other factors that may create spurious correlations within the datasets. We make the assumption that these factors do not significantly change the underlying causal relationships between gene expression and overall survival (OS), yet that they shift the distribution of gene expression profiles. The ability to perform well on data from unseen domains is defined as domain generalization [7]. Secondly, there may be shifts in the distribution of specific subgroups of breast cancer between training and test cohorts. Breast cancer tumors are routinely subtyped based on hormone status and human epidermal growth factor receptor 2 (HER2) amplification [8] with these subtypes showing differential survival outcomes [9]. Differences in subtype distributions between cohorts may lead to poor domain generalization if, for instance, the DNN primarily bases decisions on relationships between genes and OS valid only for the most prevalent subtypes in the training set.

The aim of this study is to evaluate domain and subgroup shift methods in the context of breast cancer prognosis from RNA-sequencing (RNA-seq) based gene expression profiles. Specifically, we formulate and test the following three hypotheses in relation to breast cancer prognosis from gene expression: (1) Performance of DNN models trained with empirical risk minimization (ERM) deteriorate when underlying distributions shift. (2) Methods aimed to address robustness to domain shift can improve OOD generalization. (3) Explicitly addressing shifts in clinical subtypes improves OOD generalization. We restrict this study to the scenario where the learning algorithms only have access to one training dataset but their generalization across other domains is evaluated.

II. Related Work

Robustness of DNNs to distribution shifts has become an active area of research due to its centrality to making these models applicable in real-world scenarios [3]. Common benchmarks for developing such methods have focused on image and language datasets, ranging from the magnitude of 10^4 to 10^5 in sample size and often with classification as the target [10]. Generally, methods have not been specifically developed for or evaluated on molecular data, where most datasets are of significantly smaller sample size, and neither in the context of survival analysis.

For biomedical applications, methods for improved domain generalization have been applied to computational pathology, mirroring the aforementioned focus on natural images [11]. Few studies apply robustness methods to molecular and survival data. Zhong et al. [12] increase domain generalization by applying risk extrapolation [13] to predicting malignant cell annotation from single cell RNA-seq and spatial transcriptomic data. This differs to our study in the type of input data, task, and requirement of access to multiple domains during training. Most closely related to our work is Pfisterer et al. [14], who evaluate domain generalization algorithms on predicting time to death of lymphoma and lung cancer patients from selected demographic, clinical, and lab test covariates. This study differs from ours in that input data are low-dimensional and non-molecular, meaning that it potentially adheres to different sources of distribution shift. Neither study evaluates methods explicitly addressing subgroup shifts.

III. Problem Definition

In the following, we formalize the notion of IID and OOD performance and the idea of distribution and subgroup shifts in relation to breast cancer survival prediction. We define $\mathcal{X}$ as the feature space of gene expression count values generated from RNA-seq technology. Furthermore, we define label space $\mathcal{Y}$ as an event indicator δ denoting whether an event has occurred during the follow-up period of the study and a time variable t which records the days to the event or, in the case of censored data, time to the last follow-up. Hence, a patient i is described as $\{x_i, \delta_i, t_i\}$.

The aim of ERM is to find a function $f_{\theta_{ERM}} : \mathcal{X} \to \mathcal{Y}$ that minimizes the error on a test set drawn from the distribution P_{test} by minimizing the error on a training set drawn from P_{train}. In order for this generalization to hold, training and test samples must be drawn IID i.e., $P_{train} = P_{test}$. However, in practice this assumption is often violated where $P_{train} \neq P_{test}$, which defines the OOD setting.

We consider different datasets, each collected under different condition $e \in \mathcal{E}$. While all considered datasets consist of breast cancer survival data with $y_e \in \mathcal{Y}$ and gene expression input data extracted from the primary tumor $x_e \in \mathcal{X}$, conditions vary due to geographic location, treatment strategies, sampling bias towards a subpopulation of breast cancer patients, and so on. It is assumed that each dataset is sampled IID from P_e but that distributions between conditions vary in $P_e(X)$, $P_e(Y|X)$, or both. Crucially, we make the assumption that there exists a subset of features S^*, that are causally linked to the OS and that these are largely the same across patient cohorts.

As a distinction from other studies [12], we restrict our study to the scenario where only one dataset is available during training and model selection. To only have access to one or a few training dataset under different conditions when developing machine learning models for biomedical applications is a realistic scenario. Thus, we exclude methods that require datasets under multiple conditions during training and/or validation.

The learning algorithms included in this study are evaluated in how well they minimize the error on the external test sets,

$$\min_{\theta} R_{OOD}(f_\theta) = \frac{1}{m} \sum_{e \neq e^*}^{m} R_e(f_\theta), \tag{1}$$

where R_e is the error on dataset e when performing prediction, e^* is the training set, m is the number of external test sets, and θ are trainable parameters of the DNN. $R_{OOD}(f_\theta)$ can then be compared between different methods.

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From a learning perspective, a well generalizing model bases its predictions on a set of features $S \subseteq S^*$ as these are not effected by changes in conditions. Conversely, a classifier based on features $S^- \notin S^*$, i.e., based on spurious correlation in the training set, will generalize poorly since these correlations likely will not be present in the external test sets.

In the context of subgroup shift, we assume that the training distribution $P_{train} = \sum_{g \in \mathcal{G}} \alpha_g P_g$ where $g \in \mathcal{G}$ represents different groups, in our case clinical subtypes of breast cancer, α_g is the proportion of group g, and P_g is the joint distribution over gene expression values and survival time for g. Thus, this means that the training distribution is a mixture of group distributions. Let $P_{test} = \sum_{g \in \mathcal{G}} \beta_g P_g$ where $\beta_g \approx \alpha_g$ in the IID setting and $\beta_g \neq \alpha_g$ under subgroup shift [15]. In the case of breast cancer prognosis, we expect the main challenge introduced by subgroup shifts to be that of imbalance between subtypes which may lead models to primarily learn to predict majority groups in the training distribution well. Additionally, as subtypes show differential survival outcomes, namely $P_g(Y) \neq P_{g'}(Y)$, shifts in the subpopulation also introduce label shifts which may further reduce generalizability of the learned model.

IV. Experimental Setup

A. Endpoints

The aim of this study is to evaluate domain- and subgroup-shift methods in the realm of time-to-event (TTE) prediction. However, in cancer studies generally, and in breast cancer specifically, different endpoints are measured to, for instance, assess treatment efficacy. In this study, we adopt OS as our main endpoint, defined as time from diagnosis or randomization in a randomized controlled trial to death of the patient. OS includes death from all causes not just breast cancer-specific [16, 17].

Since measuring OS for a large part of the sample may require long follow-up periods, especially when survival rates are high, other endpoints are often measured. To evaluate the domain generalization methods on more external test sets, we also include another TTE endpoint. Recurrence-free interval (RFi) is defined as death from breast cancer or recurrence/progression in the ipsilateral breast [17]. We use this endpoint for learning from the training set when making predictions for recurrence related endpoints on downstream datasets.

B. Subgroups

Breast cancer is a heterogeneous disease with different mechanisms underlying its progression and thereby patients' prognosis. Clinically, tumors are subtyped by their estrogen (ER) and progesterone receptor (PR), as well as HER2 status because these are readily determined through immunohistochemistry and can inform treatment strategies [8]. Based on these markers, we follow the definition in [18] that defines the subgroups as the combinations of ER and HER2 status. The largest subtype, ER positive- HER2 negative (ERpHER2n), is further subdivided by lymph node status (LN). Triple negative breast cancer (TNBC) is a subtype defined by the absence of positive hormone status and lack of HER2 overexpression.

C. Datasets

The datasets included in this study are described in Table I. As source training dataset, we utilize the population-based study by the *Sweden Cancerome Analysis Network-Breast (SCAN-B)* [18]. Patients were sampled in the southern region of Sweden, where all patients diagnosed with breast cancer are offered to participate in the study. Thus, no sampling bias towards any subpopulation of breast cancer e.g., clinical subtype, has been made resulting in a subtype distribution approaching that in the breast cancer population. From this cohort, we include 7,598 patients with gene expression profiles generated with RNA-seq and OS information. In contrast to the original study [18], we create a combined dataset from their training and test sets, ensuring that there are no duplications of patients.

TABLE I
Description of Cohorts included in the study

Cohort	SCAN-B	TCGA BRCA	AURORA US	GSE 177043	OEP 000155
n	7,598	1,033	44	41	360
Events	1,225	103	36	15	49
Max follow up	4,337	7,067	11,713	9,554	3,320
ERpHER2n	78.6%	40.6%	29.5%	0%	0%
HER2pERp	8.2%	10.7%	6.8%	0%	0%
HER2pERn	3.8%	3.3%	9.1%	0%	0%
TNBC	9.4%	10.9%	36.6%	100%	100%
Unknown	0%	35.5%	20.1%	0%	0%
Conserv. Prepro.	FT FPKM	FF/FFPE FPKM	FF/FFPE FPKM	FF/FFPE geTMM	FF FPKM

We include four external test sets to evaluate the domain generalization of different models. *The Cancer Genome Atlas-BRCA (TCGA-BRCA)* study [19] consists of 1,033 breast cancer patients of various subtypes with OS information and RNA-seq gene expression profiles. The subtype distribution seems to be similar to that of SCAN-B while there is a large proportion of unknown subtype.

The *AURORA-US* study [20] consists of 44 metastatic breast cancer patients with OS and RNA-seq information. This study is different in its mean OS time and frequency of event. It is well known that breast cancer in the metastatic stage has a much lower survival rate. The clinical subtypes distribution is also different from SCAN-B since proportion of cases resulting in late stage disease differ by subtype.

The study *GSE177043* [21], identified by its GEO accession number, consists of 41 node negative TNBC with OS and RNA-seq information. It is part of a larger study conducted in the Netherlands, investigating spatial immunophenotypes of TNBC. Completely consisting of this subtype, this dataset constitutes a strong subgroup shift from the training distribution.

Similarly, the Chinese study *OEP000155* consists of only TNBC subtype. The cohort is with 360 included patient cases significantly larger [22]. The primary endpoint of the study

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was relapse-free survival (RFS) defined by the authors as time from diagnosis to first recurrence or contra lateral disease [23]. This somewhat diverges from the RFS definition in [17], which excludes contra lateral disease diagnosis, and it is unclear which causes of death are included. We pretrain the recurrence model on SCAN-B with RFi as there are most uncensored patients available.

D. Evaluation

To evaluate the described methods, we use the concordance index (C-index) [24] as performance metric. Briefly, the C-index measures the probability for a pair of patients to be correctly ordered, such that a patient with shorter survival time is assigned a higher hazard by the model. For subgroup analysis, we also report the *group gap* defined as the difference between the worst-group and best-group C-index. For subgroup performance, we report the within-subgroup C-index. The reported metrics result from ten repeated runs with different random seeds. We report mean and standard deviation of the runs.

The SCAN-B dataset was randomly split into 80% training, 10% validation, and 10% test set. Model selection was done with 50 iterations of random search over the hyperparameter space chosen based on the validation set. Note that we use the mean across samples C-index for model selection. We have performed analysis with worst-group loss for model selection but could not find any apparent difference (data not shown). Before predicting on the external datasets, the final model was refitted on the entire SCAN-B dataset.

V. Methods

All models that we evaluate are fully-connected DNNs with batch normalization [25] and dropout [26] layers, trained based on the negative log partial likelihood loss (NLPLLoss) [27] commonly used in survival analysis. Briefly, a given survival model $f_\theta(x)$ outputs a so-called hazard $\hat{h}_\theta \in \mathbb{R}$. The NLPLLoss is formulated as follows,

$$\mathcal{L}_{surv} = -\frac{1}{N_{\delta=1}} \sum_{\delta_i=1} \left(\hat{h}_\theta(x_i) - log \sum_{j \in \mathcal{R}(T_i)} e^{\hat{h}_\theta(x_j)} \right), \quad (2)$$

where N is the sample size of a mini-batch and $\mathcal{R}$ is the risk set defined as all samples in the mini-batch that have lived beyond survival time T_i of patient i. Intuitively, the loss function compares $\hat{h}(x_i)$ for patients that experienced the event with all patients that have outlived patient i, including those that are censored.

A. ERM Baseline

To train the *ERM baseline* model, we learn θ_{ERM} by minimizing the $\mathcal{L}_{surv}$ over the training set consisting of all clinical subtypes, i.e the full SCAN-B training dataset. Additionally, we train five separate models on the subsets of training samples belonging to the different clinical subtypes. We refer to these models collectively as *ERM-Subtype*. This allows us to test whether it is beneficial to train pan-subtype models over expert models for each subtype. There might be a large overlap between the causal features shared between subtypes that pan-subtype models can take advantage of.

B. Methods Addressing Distribution Shift

In *multiple episode learning (MEL)* [6], n models are trained with the exact same architecture, hyperparameters and input data, with the only difference being different random seeds for weight initialization. We evaluate the method with $n \in \{2, 8, 16\}$. This simple, yet computationally expensive, approach can learn richer representations that generalize better when a linear classifier is trained on the concatenated representations from the models. This is due to different initializations resulting in different representations, containing complementary information. Thereby, the chance of including features of S^* is increased leading to higher OOD generalization. While, in contrast to ensemble methods, no diversity of representations is explicitly engineered, the simplicity of the method leads us to test its effect on the OOD generalization of survival model for breast cancer prognosis.

Additionally, we evaluate *stochastic weight averaging (SWA)* [28], a method that collects multiple sets of weights along the trajectory of stochastic gradient descent (SGD) during training. These sets of weights are then averaged in a final model. SGD-based algorithms tend to end at local minima that are close to steep ascents. Moreover, when finding a basin of low training loss, these algorithms tend to converge at the edge of such area. Thus, small shifts in loss landscape between training and test set can lead to a strong decrease in generalization. By collecting multiple model sets of weights around the low training loss basin and averaging those, it is possible to obtain final weights located centrally within the basin, making the model less sensitive to shifts between training and test set.

To explore diverse model weights around the local low loss basin, SWA either keeps a constant learning rate (*SWA-constant*) or a cyclical learning rate (*SWA-cycle*) which collects weights at the end of each cycle of a decaying learning rate. We evaluate both these learning rate schedules in this study.

C. Methods Addressing Subgroup Shift

Large differences in the distribution between different subgroups may hurt OOD generalization if the model learns to primarily base its predictions on features that correlate with the majority groups. For instance, under the ERM paradigm, the loss of all samples is summed, giving larger importance on larger groups. Rebalancing the data to make group sizes more similar to each other has thus been proposed and shown to improve worst-group performance [29]. We therefore evaluate a downsampling scheme where the majority classes are randomly subsampled. Under the *subsampling-low (SUBL)* method all groups are subsampled such that they have the same size as the smallest group. In the *subsampling-median (SUBM)* all groups larger than the median group are downsampled to the median. The groups below the median group remain the same size.

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A different, commonly applied, method to address differences in subgroup performance, and thereby implicitly robustness to subgroup shifts, is *group Distributionally Robust Optimization (group DRO)* [30]. In group DRO, instead of minimizing the average performance over the training set, or mini-batch, the empirical worst group loss (WGL) is minimized. Formally,

$$\hat{\theta}_{gDRO} := \arg\min_{\theta_{gDRO}} \max_{g \in \mathcal{G}} \{R_g(f_\theta(x), y)\}, \tag{3}$$

where R_g is the $\mathcal{L}_{surv}$ loss over each group. As demonstrated by Sagawa et al. [30], to generalize well, group DRO requires strong regularization. Therefore, we are implementing these methods with an increased weight decay. It has also been demonstrated that despite strong regularization, smaller groups tend to have a large gap between their training and test performance. To address this, [30] propose to account for group size by adding a scaling factor to problem 3,

$$\hat{\theta}_{adj} := \arg\min_{\theta_{adj}} \max_{g \in \mathcal{G}} \left\{ R_g(f_\theta(x), y) + \frac{C}{\sqrt{n_g}} \right\}, \tag{4}$$

where C is a hyperparameter representing model capacity and n_g is the group size of group g. This method is referred to as *group-adjusted DRO*.

D. Implementation

As preprocessing steps, SCAN-B and OEP00155 were normalized to fragments per kilobase of transcript per million mapped reads (fpkm) values in the original studies. We transformed raw counts of AURORA-US to fpkm values with the *R* package *countToFPKM*. For GSE177043 only Gene length corrected trimmed mean of M-values (geTMM) [31] normalized values were available. All values were log2 transformed. Before standardization, p genes were selected through variance selection, where p is a hyperparameter. *Sci-kit survival version 0.22.2* was used for calculating C-index. All models were trained with *PyTorch version 2.1.1* on a NVIDIA Tesla A40 GPU. The models were optimized with Adam [32]. For SWA models, we set the search space for the start of model collection to a rather large range between 20 to 200 optimization steps, since we are not initializing the model with pretrained weights. Model depth and width, dropout, weight decay and learning rate have been determined through random search.

VI. Results and Discussion

A. Overall and Worst Group IID Performance Is Not Substantially Improved by non-ERM Methods for OS

Results displayed in Table II show that the overall IID performance when predicting OS is similar between ERM and non-ERM methods (maximum increase of 0.8 percentage points [pp]). Additionally, even methods designed to improve balanced performance on different subgroups do not strongly improve the worst group performance. This indicates that differentiating patients within the groups by their risk may be a harder task for different subtypes and not sufficiently addressed by the methods. Interestingly, DRO-based and MEL methods seem to improve the performance on the already well performing subtypes ERpHER2nLNp and HER2pERp. Furthermore, the gap between best and worst performing group is not reduced substantially with the exception of SUBL/M which however is driven by a lower worst group performance.

The worst group performance of all methods with the exception of SUBM, is observed on HER2pERn or TNBC. We conjecture that in the case of the former group this is due to its small sample size (Table I) and its lack of shared receptor status with the larger ER-positive subtypes. Similarly, although of larger sample size, the TNBC group does not share receptor status with any other group. This may make finding shared OS-related features between the groups more difficult.

B. Overall and Worst Group IID Performance Can be Improved by non-ERM Methods for RFi

For RFi, non-ERM methods improve overall performance with MEL-8 showing a 3.2pp increase in performance (see Table III). The best improvement of worst group performance can be found between group DRO and ERM with 4.9pp compared to 1.5pp for OS (between ERM and MEL-8) indicating that subgroup-specific methods can have positive effects. However, these results are to be taken with caution due to their large standard deviation. Compared to the IID results on OS, larger group gaps can be observed, driven by lower worst group performance. Finally, also when predicting RFi, worst group performance can be observed on HER2pERn and TNBC.

C. A Shared ERM Model Outperforms Subtype Expert Models

Expert models (ERM-Subtype) perform worse than a pan-subtype model (ERM) on all subtypes and endpoints (Table II and III). It can be observed that the performance of the expert models only approaches that of ERM on subgroups with large sample size (ERpHER2n). We conjecture that the superiority of training a pan-subtype model stems from shared molecular mechanisms for OS and RFi between patients with primary tumors of different subtype. Training a shared model thus allows the sharing of information between subgroups and increase sample size.

D. Non-ERM Methods Can Improve Domain Generalization with Varying Effect Sizes

We can observe that all methods result in decreased performance when predicting on all external test sets compared to the IID setting, with the exception of TCGA-BRCA (see Table IV). However, all methods except SUBL and MEL-2 show an increased mean domain generalization (R_{OOD}) across the external datasets compared to ERM, albeit to varying degrees. SWA-cycle shows the largest improvement with 2.2pp followed by MEL-16 and group-adjused DRO with both 1.5pp increase. Also, it can be observed that MEL-8, MEL-16, SWA-constant, and SWA-cycle consistently perform better than ERM on all external datasets, noting that differences can be small. None of the methods consistently outperforms all others

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TABLE II
WITHIN-SUBTYPE PERFORMANCE OF OVERALL SURVIVAL ON SCAN-B COHORT. MEAN PERFORMANCE AND STANDARD DEVIATION. BEST COLUMN PERFORMANCE IN BOLD. WORST-SUBGROUP PERFORMANCE OF EACH ALGORITHM UNDERLINED.

Algorithm	ERpHER2nLNn	ERpHER2nLNp	HER2pERp	HER2pERn	TNBC	Group Gap	Overall
ERM	0.684 (0.030)	0.731 (0.043)	0.724 (0.092)	0.656 (0.125)	0.663 (0.060)	0.075	0.718 (0.035)
ERM-Subtype	0.664 (0.031)	0.679 (0.060)	0.579 (0.096)	0.545 (0.101)	0.569 (0.078)	-	-
SUBL	0.642 (0.039)	0.678 (0.053)	0.633 (0.112)	0.619 (0.093)	0.631 (0.060)	**0.059**	0.651 (0.049)
SUBM	0.651 (0.038)	0.714 (0.022)	0.688 (0.091)	0.680 (0.050)	0.655 (0.041)	0.062	0.688 (0.027)
Group DRO	**0.707 (0.026)**	0.745 (0.046)	0.738 (0.106)	**0.711 (0.094)**	0.667 (0.044)	0.078	**0.726 (0.018)**
Group-adj. DRO	0.687 (0.046)	0.748 (0.035)	0.718 (0.011)	0.676 (0.087)	0.666 (0.043)	0.082	0.715 (0.035)
MEL2	0.695 (0.030)	**0.754 (0.041)**	**0.751 (0.075)**	0.657 (0.120)	0.662 (0.042)	0.097	0.720 (0.023)
MEL8	0.697 (0.034)	0.750 (0.026)	0.741 (0.100)	0.684 (0.097)	**0.671 (0.049)**	0.079	0.722 (0.020)
MEL16	0.694 (0.028)	0.751 (0.051)	0.748 (0.080)	0.697 (0.093)	0.660 (0.055)	0.091	0.722 (0.022)
SWA-const.	0.701 (0.027)	0.738 (0.029)	0.730 (0.095)	0.684 (0.095)	0.664 (0.048)	0.075	0.721 (0.019)
SWA-cycle	0.685 (0.039)	0.741 (0.022)	0.740 (0.064)	0.691 (0.078)	0.671 (0.070)	0.070	0.715 (0.022)

across the external test sets. From these results, we conclude that the evaluated methods can improve domain generalization compared to ERM although it is not obvious which method works best.

E. Performance Differs Between Datasets and TNBC-Dominated Cohorts Show Largest Generalization Gap

Considering TCGA-BRCA specifically, DRO-, MEL-, and SWA-based methods, with the exception of MEL-2, generalize better to this cohort than ERM (see Table IV). Moreover, all evaluated methods, again except MEL-2, perform better on TCGA-BRCA than in the IID setting on SCAN-B. This indicates that the TCGA-BRCA cohort is easier to correctly order according to patients' risk than in SCAN-B. Interestingly, despite the, most likely, similar subgroup distribution between the two cohorts, group-adjusted DRO performs best.

On the contrary, the AURORA-US cohort contains an increased proportion of TNBC patients, yet subgroup shift methods do not perform better than ERM. The main difference between the cohorts is that SCAN-B is dominated by breast cancer in early stage, while AURORA-US explicitly samples metastatic breast cancer patients. We conjecture that MEL- and SWA-based methods are able to capture features in the training set that are less idiosyncratic to early stage breast cancer compared to SUB- and DRO-based methods.

Finally, on purely TNBC containing cohorts the methods do not generalize well. On GSE177043, generalization is unstable with large standard deviation of performance between runs. Generalization to OEP000155 is the worst among the external test set with all methods performing worse than a C-index of 0.60. We hypothesize three potential reasons for this large generalization gap. Firstly, the definitions of recurrence based endpoints used to pretrain on the SCAN-B and evaluate on OEP000155 do not fully align and how closely they match is unknown. Secondly, the aforementioned poor IID generalization to the TNBC subgroup may lead to the poor OOD generalization. Thirdly, TNBC is a highly heterogeneous subgroup with TNBC subtypes in turn [33]. This may lead to stronger shifts between cohorts. Regardless the reason, it is important to outline that TNBC with its worse prognosis and limited targeted treatment options, constitutes a larger unmet clinical need compared to other subtypes [9, 33]. This highlights the need for further research into learning more robust models for TNBC.

VII. CONCLUSION

The results of this study show that ERM shows a generalization gap for most domains, partially confirming hypothesis 1. From evaluating current methods for addressing domain generalization and specifically subgroup shifts on the task of breast cancer prognosis, we conclude that there are varying degrees of generalization gaps between cohorts. The evaluated methods are able to narrow these gaps to some degree. However, we also highlight that generalization to some of the cohorts, especially those dominated by TNBC samples, remains poor. Thus, we provide evidence that these methods can address shifts in distributions but that the degradation in performance is not fully addressed and differences to ERM can be small (hypothesis 2). Interestingly, we find no evidence that subgroup shift specific methods perform better than domain generalization methods on cohorts with strong subgroup shifts (hypothesis 3). However, they do improve over ERM slightly.

Overall, we conclude that some of the evaluated method can reduce the generalization gap even on vastly different data types and tasks compared to those on which they were developed. Yet, we also emphasize that methods are not consistent in their improvements to OOD datasets, since there is not one method superior under all shifts. This is in line with previous research [14].

Analyzing the IID subgroup performance indicates that TNBC may be the most challenging subtype to make prognosis for. Therefore, we highlight the need to further investigate methods improving survival and recurrence analysis for TNBC.

This study is limited to evaluating domain generalization across the same endpoints. It remains an open question whether the methods studied also improve transfer learning to other endpoints. Additionally, the methods were developed mostly in the context of natural images and language, aimed at classification tasks. Robust survival analysis methods in the deep learning context are still an unmet need.

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TABLE III
WITHIN-SUBTYPE PERFORMANCE OF RECURRENCE FREE INTERVAL ON SCAN-B COHORT. MEAN PERFORMANCE AND STANDARD DEVIATION. BEST COLUMN PERFORMANCE IN BOLD. WORST-SUBGROUP PERFORMANCE OF EACH ALGORITHM UNDERLINED.

Algorithm	ERpHER2nLNn	ERpHER2nLNp	HER2pERp	HER2pERn	TNBC	Group Gap	Overall
ERM	0.737 (0.068)	0.700 (0.054)	0.716 (0.154)	0.597 (0.218)	**0.687 (0.104)**	0.140	0.739 (0.026)
ERM-Subtype	0.705 (0.069)	0.657 (0.068)	0.574 (0.149)	0.555 (0.195)	0.559 (0.080)	-	-
SUBL	0.656 (0.068)	0.680 (0.078)	0.693 (0.115)	0.592 (0.146)	0.606 (0.112)	0.101	0.684 (0.041)
SUBM	0.730 (0.050	0.699 (0.077)	0.698 (0.121)	**0.700 (0.135)**	0.618 (0.096)	0.112	0.734 (0.031)
Group DRO	0.714 (0.065)	0.724 (0.098)	0.740 (0.129)	0.682 (0.108)	0.646 (0.131)	0.094	0.748 (0.024)
Group-adj. DRO	0.722 (0.064)	0.743 (0.074)	**0.752 (0.114)**	0.648 (0.142)	0.624 (0.100)	0.128	0.751 (0.023)
MEL2	0.755 (0.080)	0.747 (0.052)	0.744 (0.079)	0.620 (0.196)	0.666 (0.146)	0.135	0.767 (0.033)
MEL8	**0.758 (0.076)**	**0.756 (0.059)**	0.747 (0.079)	0.642 (0.210)	0.630 (0.152)	0.128	**0.771 (0.027)**
MEL16	0.736 (0.094)	0.738 (0.069)	0.735 (0.097)	0.612 (0.168)	0.663 (0.138)	0.126	0.762 (0.052)
SWA-const.	0.718 (0.058)	0.727 (0.053)	0.728 (0.090)	0.699 (0.094)	0.637 (0.097)	**0.091**	0.755 (0.029)
SWA-cycle	0.725 (0.056)	0.732 (0.071)	0.749 (0.102)	0.697 (0.100)	0.657 (0.094)	0.092	0.755 (0.023)

TABLE IV
DOMAIN GENERALIZATION PERFORMANCE. MEAN PERFORMANCE AND STANDARD DEVIATION. BEST PERFORAMNCE IN BOLD

Algorithm	TCGA-BRCA	AURORA-US	GSE177043	OEP000155	R_{OOD}
ERM	0.718 (0.035)	0.629 (0.015)	0.615 (0.093)	0.581 (0.049)	0.636
SUBL	0.697 (0.036)	0.611 (0.058)	**0.655 (0.107)**	0.556 (0.043)	0.630
SUBM	0.709 (0.026)	0.625 (0.046)	0.641 (0.079)	0.575 (0.029)	0.641
Group DRO	0.739 (0.027)	0.622 (0.055)	0.636 (0.111)	0.588 (0.036)	0.646
Group-adj. DRO	**0.749 (0.010)**	0.617 (0.049)	0.640 (0.081)	**0.599 (0.019)**	0.651
MEL-2	0.714 (0.028)	0.646 (0.034)	0.595 (0.059)	0.573 (0.035)	0.632
MEL-8	0.735 (0.022)	0.648 (0.037)	0.647 (0.059)	0.592 (0.028)	0.656
MEL-16	0.738 (0.014)	0.654 (0.032)	0.619 (0.052)	0.592 (0.015)	0.651
SWA-const.	0.742 (0.012)	0.649 (0.020)	0.619 (0.070)	0.586 (0.023)	0.649
SWA-cycle	0.743 (0.012)	**0.655 (0.022)**	0.639 (0.064)	0.596 (0.013)	**0.658**

VIII. DATA AVAILABILITY

RNA-seq values and clinical information of SCAN-B are available through the original study [18]. TCGA-BRCA clinical and expression data can be found at Genomic Data Commons Data Portal https://portal.gdc.cancer.gov/. For AURORA-US, RNA-seq values can be found under GEO accession number *GSE209998* and clinical information in [20]. RNA-seq and clinical data of GSE177043 can be accessed under the accession number and in the, supplementary files of [21], respectively. OEP000155 data is available through [22].

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